

Image Processing & Recognition: Lecture 6 Review

Welcome to Level 6! We are moving beyond simple classification ("Is this a cat?") to **Object Detection** ("Where is the cat?").

Part 1: What are you learning? (The "Human" Summary)

1. The Pre-Game: Better Data

- **Image Augmentation:** To train robust models, we fake new data by flipping, cropping, and changing the color of our existing images. This stops the model from memorizing exact pixel values.
- **Fine-Tuning:** Instead of training a massive brain from scratch, we take a pre-trained genius (like a model trained on ImageNet) and "teach" it a new specific task (Transfer Learning).

2. The Core Concept: Bounding Boxes

- **Bounding Box:** A rectangle $[x_{min}, y_{min}, x_{max}, y_{max}]$ that wraps around an object.
- **Anchor Boxes:** The model doesn't just guess a box from scratch. It generates thousands of pre-defined "guesses" (Anchor Boxes) of different sizes and aspect ratios all over the image, then checks if any of them fit an object.

3. The "Judge": Intersection over Union (IoU)

How do we know if a box is good?

- **IoU:** We calculate the area where two boxes overlap (Intersection) and divide it by the total area covered by both (Union).
 - $\text{IoU} = 1.0 \rightarrow$ Perfect match.
 - $\text{IoU} = 0.0 \rightarrow$ No overlap.
 - $\text{IoU} > 0.5 \rightarrow$ Usually considered a "good detection."

4. The Models (Evolution of Detection)

- **R-CNN:** The slow grandpa. It extracts 2000 regions and runs a CNN on *each one*. Very slow.
- **Fast R-CNN:** Smarter. Runs the CNN *once* on the whole image, then picks regions from the feature map.
- **Faster R-CNN:** The standard. Replaces the slow region selection method with a neural network (Region Proposal Network).
- **SSD (Single Shot Detector):** The speed demon. It detects objects in one go (no separate region proposal step) by looking at feature maps of different scales.

Part 2: The Solved Problem (Step-by-Step)

Here is a formal problem statement modeled after the exercises in the previous course PDFs.

Exercise 6

Assume we have an image where the ground-truth bounding box for an object, denoted as B_{gt} , and a predicted bounding box, denoted as B_p , are defined by the coordinates of their upper-left corner (x_{min}, y_{min}) and lower-right corner (x_{max}, y_{max}) .

Given: $B_{gt} = [50, 50, 150, 150]$ $B_p = [100, 100, 200, 200]$

Compute the Intersection over Union (IoU) to quantify the similarity between the predicted box and the ground truth.

Solution:

Step 1: Understand the Bounding Box Definition We define the boxes using the coordinate system where the origin is the upper-left corner of the image.

- **Source: Page 22**, which states: "The bounding box is rectangular, which is determined by the x and y coordinates of the upper-left corner... and... lower-right corner."

Step 2: Calculate the Area of Each Box The area of a box is $(x_{max} - x_{min}) \times (y_{max} - y_{min})$.

- Area of Ground Truth ($Area_{gt}$): $(150 - 50) \times (150 - 50) = 100 \times 100 = 10,000$ pixels.
- Area of Prediction ($Area_p$): $(200 - 100) \times (200 - 100) = 100 \times 100 = 10,000$ pixels.

Step 3: Calculate the Intersection Area ($|A \cap B|$) We must find the coordinates of the overlapping rectangle.

- $x_{start} = \max(x_{min}^{gt}, x_{min}^p) = \max(50, 100) = 100$
- $y_{start} = \max(y_{min}^{gt}, y_{min}^p) = \max(50, 100) = 100$
- $x_{end} = \min(x_{max}^{gt}, x_{max}^p) = \min(150, 200) = 150$
- $y_{end} = \min(y_{max}^{gt}, y_{max}^p) = \min(150, 200) = 150$
- Intersection Area = $(150 - 100) \times (150 - 100) = 50 \times 50 = 2,500$.
- **Source: Page 28** (Jaccard Index definition) and **Page 30** (Figure 2 showing the "Intersection" region).

Step 4: Calculate the Union Area ($|A \cup B|$) The union is the total area covered by both boxes combined.

- Formula: $|A \cup B| = |A| + |B| - |A \cap B|$
- $Union = 10,000 + 10,000 - 2,500 = 17,500$.
- **Source: Page 28** (Jaccard Index formula denominator) and **Page 30** (Figure 2 showing the "Union" region).

Step 5: Compute IoU Finally, we compute the ratio.

- Formula: $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$
- $IoU = \frac{2,500}{17,500} = \frac{1}{7} \approx 0.143$.
- **Source: Page 28** (Jaccard Index formula) and **Page 30** (IoU formula).

Part 3: Your Turn (Interactive Exercises)

Practice Problem 1: The Perfect Overlap?

Let's test a case with better alignment.

Setup:

- **Box A:** $[0, 0, 10, 10]$ ($x_{min}, y_{min}, x_{max}, y_{max}$).
- **Box B:** $[0, 0, 5, 10]$.

Task: Calculate the IoU.

Solving Explanation:

1. **Areas:**
 - Area A: $(10 - 0) \times (10 - 0) = 100$.
 - Area B: $(5 - 0) \times (10 - 0) = 50$.
2. **Intersection:**
 - x -overlap: 0 to 5 (width 5).
 - y -overlap: 0 to 10 (height 10).
 - Intersection Area: $5 \times 10 = 50$.
3. **Union:**
 - $100 + 50 - 50 = 100$.
4. **IoU:**
 - $50/100 = 0.5$.

Result: IoU is **0.5**. (Box B is completely inside Box A, covering half its area).

Practice Problem 2: No Overlap

Setup:

- **Box A:** $[0, 0, 2, 2]$.
- **Box B:** $[3, 3, 5, 5]$.

Task: Calculate the IoU.

Solving Explanation:

1. **Areas:**

- Area A: $2 \times 2 = 4$.
- Area B: $2 \times 2 = 4$.

2. **Intersection:**

- $x_{start} = \max(0, 3) = 3$.
- $x_{end} = \min(2, 5) = 2$.
- Width = $2 - 3 = -1$.
- Since width is negative, there is **no overlap**. Intersection Area = 0.

3. **Union:**

- $4 + 4 - 0 = 8$.

4. **IoU:**

- $0/8 = 0$.

Result: IoU is 0